is a potential control point at which gene expression can be turned on or off, accelerated, or slowed down. Many genes have more than one control point.

When the structure of DNA was determined in 1953, an understanding of the mechanisms that control gene expression in eukaryotes seemed almost hopelessly out of reach. Since then, advances in DNA technology (see Chapter 20) have enabled molecular biologists to uncover many details of eukaryotic gene regulation. In all organisms, gene expression is commonly controlled at transcription; regulation at this stage often occurs in response to signals coming from outside the cell, such as hormones or other signaling molecules. For this reason, the term *gene expression* is often equated with transcription for both bacteria and eukaryotes. While this may be the case for bacteria, the greater complexity of eukaryotic cell structure and function provides opportunities for regulating gene expression at many stages besides transcription (see Figure 18.6). Let's examine some of the important control points of eukaryotic gene expression more closely.

Regulation of Chromatin Structure

Recall that the DNA of eukaryotic cells is packaged with proteins in an elaborate complex known as chromatin, the basic unit of which is the nucleosome (see Figure 16.23). The structural organization of chromatin not only packs a cell's DNA into a compact form that fits inside the nucleus, but also helps regulate gene expression in several ways. Genes within heterochromatin, which is more densely arranged than euchromatin, are usually not expressed. In euchromatin, whether or not a gene is transcribed is affected by the location of nucleosomes along a gene's promoter and also the sites where the DNA attaches to the protein scaffolding of the chromosome. Chromatin structure and gene expression can be influenced by chemical modifications of both the histone proteins of the nucleosomes around which DNA is wrapped and the nucleotides that make up that DNA. Here we examine the effects of these modifications, which are catalyzed by specific enzymes.

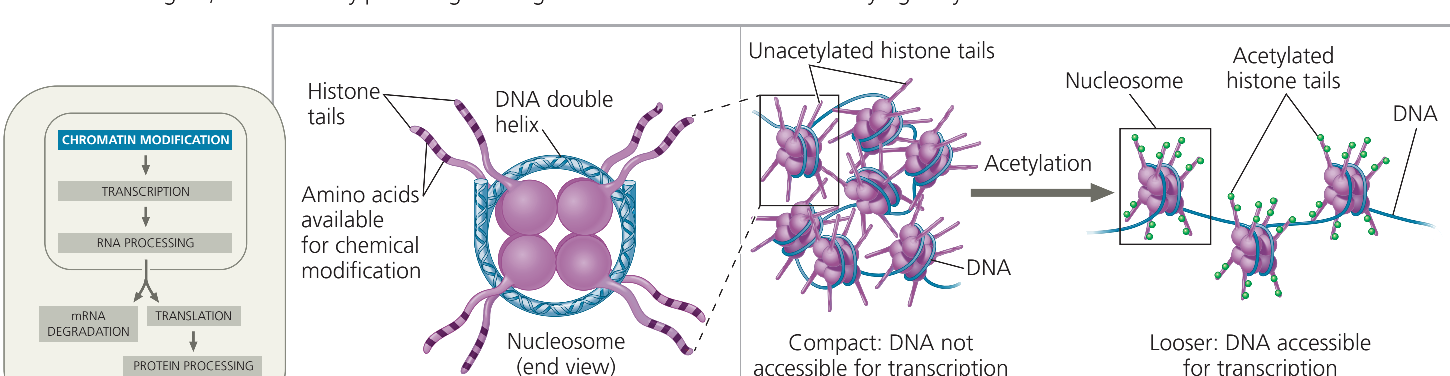
Histone Modifications and DNA Methylation

Chemical modifications to histones, found in all eukaryotic organisms, play a direct role in the regulation of gene transcription. The N-terminus of each histone protein in a nucleosome protrudes outward from the nucleosome **(Figure 18.7a)**. These so-called *histone tails* are accessible to various modifying enzymes that catalyze the addition or removal of specific chemical groups, such as acetyl (—$COCH_3$), methyl, and phosphate groups (see Figure 4.9). Generally, **histone acetylation**—the addition of an acetyl group to an amino acid in a histone tail—appears to promote transcription by opening up chromatin structure **(Figure 18.7b)**, while the addition of methyl groups to histones can lead to the condensation of chromatin and reduced transcription. Often, the addition of a particular chemical group may create a new binding site for enzymes that further modify chromatin structure.

Rather than modifying histone proteins, a different set of enzymes can methylate the DNA itself on certain bases, usually cytosine. Such **DNA methylation** occurs in most plants, animals, and fungi. Long stretches of inactive DNA, such as that of inactivated mammalian X chromosomes (see Figure 15.8), are generally more methylated than regions of actively transcribed DNA. On a smaller scale, the DNA of individual genes is usually more heavily methylated in cells in which those genes are not expressed. Removal of the extra methyl groups can turn on some of these genes.

▼ Figure 18.7 A simple model of histone tails and the effect of histone acetylation. Amino acids in histone tails may be chemically modified by addition of acetyl groups (green balls) or other groups (such as methyl or phosphate groups). Such modifications affect chromatin structure in a region, sometimes by providing binding sites for other chromatin-modifying enzymes.

Mastering Biology HHMI Animation: Role of MECP2 in Gene Silencing



(a) Histone tails protrude outward from a nucleosome. The amino acids in the histone tails are accessible for chemical modification.

(b) Acetylation of histone tails promotes loose chromatin structure that permits transcription. A region of chromatin in which nucleosomes are unacetylated forms a more condensed arrangement of chromatin (left) in which the DNA is not transcribed. When nucleosomes become highly acetylated (right), the chromatin becomes less condensed, and the DNA is accessible for transcription.